

ICT2142 E- Business Systems

Lecture 01

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Introduction

- Course Unit
 - ICT2142 E-Business Systems
- Credits – 2
- Lecture Hours – 45h (15h – Theory, 30h – Practical)

Aim of the Module



To provide a comprehensive understanding of modern e-business environments by exploring digital business models, technologies, and emerging trends while improving practical skills needed to analyze, design, and manage e-business solutions, and to apply digital technologies effectively.



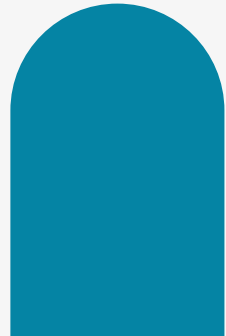
Course Content

- Introduction to E-Business and E-Commerce
- E-Business Models and Digital Value Creation
- E-Commerce Infrastructure and Technologies
- E-Commerce Payment Systems
- Designing & Developing E-Commerce Websites
- E-Commerce Platforms & Tools
- Digital Marketing for E-Business
- Data Analytics for E-Commerce
- E-Business Security and Legal Aspects
- Logistics, Fulfilment & Supply Chain in E-Business
- Emerging Trends in E-Business

Practical Work

- Technical evaluation of popular e-commerce platforms
- Building a simple prototype e-commerce store
- Basic integration with a payment gateway

References



- Electronic Commerce: A Managerial and Social Networks Perspective by Efraim Turban, David King, Jae Kyu Lee, Ting-Peng Liang, Deborrah C. Turban (9th Edition, 2018)
- Introduction to Electronic Commerce and Social Commerce by Efraim Turban, Judy Whiteside, David King, Jon Outland (4th Edition, 2017)

Evaluation Criteria

- Theory Exam: 70%
- Continuous Assessment : 30%
 - ✓ Quizzes: 10%
 - ✓ Mini Project: 20%



Introduction to Electronic Commerce



Objectives

- Evolution of electronic business
- Difference between e-business and e-commerce
- Digital business ecosystems
- Current global and Sri Lankan e-commerce trends
- SWOT Analysis

What is E-Business and E-Commerce?

- **Electronic business (e-business)**

Doing business electronically by completing business processes over open networks, thereby substituting information for physical business process.

- **Electronic commerce (e-commerce)**

E-commerce (electronic commerce) is the buying and selling of goods and services, over an electronic network, primarily the internet.

E-Business VS E-Commerce



E-Business VS E-Commerce

E- Business is a collection of

- eCommerce +
- Customer relationship management +
- Supply chain management +
- Knowledge management +
- Business intelligence +
- Collaborative technologies

BASIS FOR COMPARISON	E-COMMERCE	E-BUSINESS
Meaning	Trading of merchandise, over the internet is known as E-commerce.	Running business using the internet is known as E-business.
What is it?	Subset	Superset
Is it limited to monetary transactions?	Yes	No
What they carry out?	Commercial transactions	Business transactions
Requires	Website	Website, CRM, ERP, etc.
Which network is used?	Internet	Internet, Intranet and Extranet.

Objectives of E-Commerce

- Primarily, increasing the speed and efficiency of
 - Business transactions and processes
- Moreover,
 - Enhanced competitiveness
 - Job creation
 - Economic growth
 - Lower price for goods and services through competition
 - Streamline and much simpler business processes

Categories of E-Commerce

Six general e-commerce categories

- Business-to-Business (B2B)
- Business-to-Consumer (B2C)
- Consumer-to-Consumer (C2C)
- Consumer-to-Business (C2B).
- Business-to-Administration (B2A)
- Consumer-to-Administration (C2A)

Six Categories of E-Commerce

Category	Description	Examples
Business-to-Business (B2B)	(B2B) e-commerce encompasses all electronic transactions of goods or services conducted between companies.	Producers and traditional commerce wholesalers
Business-to-Consumer (B2C)	It corresponds to the retail section of e-commerce, where traditional retail trade normally operates.	sell all kinds of consumer goods, such as computers, software, books, shoes, cars, food, financial products, digital publications, etc.
Consumer-to-Consumer (C2C)	Consumer-to-Consumer (C2C) type e-commerce encompasses all electronic transactions of goods or services conducted between consumers.	e Bay.com

Six Categories of E-Commerce

Category	Description	Examples
Consumer-to-Business (C2B)	A large number of individuals make their services or products available for purchase for companies seeking precisely these types of services or products.	Logo creation
Business-to-Administration (B2A)	This part of e-commerce encompasses all transactions conducted online between companies and public administration.	social security, employment, legal documents and registers
Consumer-to-Administration (C2A)	all electronic transactions conducted between individuals and public administration.	distance learning, medical appointments, payment of health services



Evolution of E-Business

Key Stages of E-Business Evolution

- Pre-Internet Era (1940s–1980s)
 - The Internet Boom (1990s)
 - Maturation and Gateways (2000s)
 - Mobile and Social Commerce (2010s)
 - Intelligent Digital Ecosystems (Present)
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Evolution of E-Business

- **Pre-Internet Era (1940s–1980s):**
 - Began with basic electronic messaging during the 1948 Berlin airlift via telex.
 - Introduced Computer-to-Computer Electronic Data Interchange (EDI) in 1975 to exchange business invoices and orders securely.
- **The Internet Boom (1990s):**
 - The World Wide Web launched in 1991, bringing business online.
 - The first secure online credit card transaction took place in 1994.
 - Retail giants like Amazon and eBay launched, revolutionizing consumer shopping.
 - IBM officially popularized the term "e-business" in 1997.
- **Maturation and Gateways (2000s):**
 - Expanded rapidly after recovering from the dot-com bubble burst.
 - Secure payment platforms like PayPal made online buying trusted and mainstream

Evolution of E-Business

- **Mobile and Social Commerce (2010s):**
 - Smartphone adoption shifted traffic to mobile apps (m-commerce).
 - Social media platforms integrated direct purchasing options.
- **Intelligent Digital Ecosystems (Present):**
 - Uses cloud computing, artificial intelligence, and big data for personalized shopping.
 - Merges online and physical stores into smooth omnichannel experiences.

Advantages of E-Commerce

- Faster buying/selling procedure, as well as easy to find products.
- Buying/selling 24/7.
- More reach to customers, there is no theoretical geographic limitations.
- No need of physical company set-ups.
- Low operational costs and better quality of services.
- Easy to start and manage a business.
- Customers can easily select products from different providers without moving around physically.

Disadvantages of E-Commerce

- Perishable grocery products are much harder to sell online.
- Anyone, good or bad, can easily start a business. And there are many bad sites which eat up customers' money.
- There is no guarantee of product quality.
- No one can buy during a site crash.
- As there is minimum chance of direct customer to company interactions, customer loyalty is always on a check.
- There are many hackers who look for opportunities, and thus an ecommerce site, service, payment gateways, all are always prone to attack.



SWOT Analysis

- Analyst first looks into the business unit to identify its strengths and weaknesses
 - Analyst then reviews operating environment and identifies opportunities and threats
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SWOT Analysis

In Class Activity

Conduct a SWOT Analysis for DELL.





Thank you... !

